

Melika Baghi

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Profile

My work begins with a simple conviction: healthcare systems generate extraordinary amounts of complex, high-dimensional data, and building reliable analytical tools to act on that data has direct consequences for patients and providers. I came to ISyE through a graduate degree in **Healthcare Systems** and a thesis on hospital scheduling and patient operations. In my Ph.D. and concurrent M.S. in Statistics, I have deepened that foundation with rigorous methods in simulation, statistical learning, uncertainty-aware machine learning, and multimodal representation learning. My long-term goal is to develop interpretable, robust, and practically deployable analytics that help healthcare systems make better decisions — at the level of individual patients, clinical operations, and system design.

Research Interests

My research develops geometry-aware and uncertainty-aware learning methods for high-dimensional systems, motivated by problems in healthcare and engineering.

Reduced-order and surrogate modeling: geometry-aware reduced-order modeling on Riemannian manifolds (Grassmann, Stiefel), tensor decompositions, active subspaces, and Gaussian-process emulation for expensive simulations.

Uncertainty quantification and trustworthy prediction: conformal and selective prediction, calibration, sensitivity analysis, and reliable decision-making under missing or shifting data.

Applications: healthcare operations and scheduling, clinical decision support, and high-dimensional engineering and scientific systems.

Education

Georgia Institute of Technology

Ph.D. in Industrial & Systems Engineering — System Informatics & Control

Atlanta, GA

Aug 2023 – Present

- Advisors: Prof. Kamran Paynabar and Prof. Xiao Liu
- Dissertation: *Geometry-Aware Learning Frameworks for Reduced-Order Modeling with Uncertainty Quantification in High-Dimensional PDE Systems*

Georgia Institute of Technology

M.S. in Statistics

Atlanta, GA

Aug 2023 – May 2026

- Thesis: *Reduced-Order Modeling with Uncertainty Quantification for High-Dimensional PDE Systems in Digital Twin Applications*
- Advisors: Prof. Kamran Paynabar and Prof. Xiao Liu

Amirkabir University of Technology (Tehran Polytechnic)

M.S. in Industrial Engineering — Healthcare Systems; GPA: 4.0/4.0 (19.35/20) Oct 2021 – Jun 2023

Tehran, Iran

- Thesis: *Scheduling of Medical Tourists Visiting Medical Centers Using an Agent-Based Approach*
- Advisor: Prof. Hadi Mosadegh
- Focused on healthcare operations research, patient flow modeling, hospital resource allocation, simulation-based decision support, and health systems management

- Direct-entry admitted as an elite student (ranked 3rd of 120 applicants)

Amirkabir University of Technology (Tehran Polytechnic)
B.S. in Industrial Engineering; GPA: 3.91 / 4.0 (18.65 / 20)

Tehran, Iran
Sep 2018 – Jun 2021

Amirkabir University of Technology (Tehran Polytechnic)
B.S. in Aerospace Engineering; GPA: 3.79 / 4.0 (18.14 / 20)

Tehran, Iran
Sep 2016 – Feb 2021

- Selected as an elite student and approved to pursue a dual degree alongside Industrial Engineering (2018)

Health Systems and Biomedical AI Research

Hospital Scheduling and Operations Analytics
M.S. Thesis Research — Prof. Hadi Mosadegh

Amirkabir University of Technology
Sep 2021 – Aug 2023

- **Hybrid Simulation Model for International Patient Scheduling in a Multi-Specialty Hospital.** Developed a hybrid agent-based and discrete-event simulation (ABS+DES) model of an international patient department serving both local patients and medical tourists across five clinical specialties: cardiology, internal medicine, cosmetic surgery, paediatrics, and breast oncology. The model represented the full patient pathway, including registration, online and in-person consultation, specialist assignment, bed allocation, inpatient stay, discharge, medication adherence, early abandonment, emergency escalation, and doctor switching.
- **Design of Experiments and Factor Screening.** Conducted a 256-run two-level fractional factorial design over 16 controllable operational factors, including bed capacities by section, specialist counts, online consultation shares, bed-scheduling rules, patient-priority policies, and clinic slot intervals. Fit screening regression models ($R^2 = 0.78\text{--}0.93$) for six operational responses: medical-tourist hospital-queue waiting time, system waiting time, early dropout, emergency escalation, recovered patients, and specialist utilization.
- **Impact and Policy Implications.** The hybrid model reduced average medical-tourist hospital-queue waiting time from **13.7 days** in a DES-only baseline to **2.4 days**, while also revealing dropout and emergency-escalation patterns that a purely procedural model would suppress. Results identified bed capacity, patient-priority rules, online versus in-person channel design, and clinic slot interval as key managerial levers for hospitals serving mixed local–international patient populations.

Certified Multimodal Selective Prediction (CEMC)
Ph.D. Research — Prof. Kamran Paynabar & Prof. Xiao Liu

Georgia Institute of Technology
2025 – Present

- **Motivation.** Clinical decision-support models routinely have to act when some inputs are missing: a patient may have imaging but not labs, notes but not genomics. Naively imputing the missing data and predicting anyway can be unsafe, while always abstaining or always acquiring more data is wasteful. *Certified Evidence-Minimal Coalitions* (CEMC) asks a sharper question — when is the evidence already in hand sufficient to act on?
- **Proposed Methodology.** Developed a conformal-prediction framework that certifies a prediction from the observed modalities when imputing the missing ones would not change the model’s decision (high imputation-induced oracle agreement), and otherwise abstains and recommends acquiring additional modalities. The framework provides distribution-free coverage guarantees and is positioned explicitly against standard conformal prediction and selective classification, with an accompanying statement of what it does and does not claim.
- **Evaluation and Healthcare Relevance.** Validated on multimodal benchmarks (PolyMNIST, DVM-CAR) and targeted at a clinical multimodal setting (HAIM-MIMIC-MM), where modality-acquisition decisions carry real cost and abstention is a safer alternative to overconfident prediction under missing evidence.

- *Manuscript in preparation (AAAI).*

Reduced-Order Modeling, Tensor Methods, and Uncertainty Quantification

The following Ph.D. projects develop statistical and computational methods for reliable, scalable surrogate modeling of high-dimensional systems — methods that transfer naturally to data-scarce scientific and biomedical settings where repeated high-fidelity evaluation is expensive.

Georgia Institute of Technology

Ph.D. Research Assistant — Prof. Kamran Paynabar & Prof. Xiao Liu

Atlanta, GA

Aug 2023 – Present

- **Geometry-Aware Reduced-Order Modeling (cXGBoost).** Proposed stable injective mappings from the Grassmann manifold to a constrained Euclidean space, enabling gradient boosting tree ensembles to learn parameter-to-subspace maps for expensive high-fidelity simulations. Integrated a quadratically constrained quadratic program (QCQP) at each tree split to enforce manifold injectivity, with projected Gaussian-process emulation on the recovered subspaces. Demonstrated on CFD and PDE benchmarks. *Manuscript under review.*
- **Tensor-Structured Surrogate Modeling (Tensor ROM).** Developed a surrogate framework that preserves the multiway structure of spatiotemporal simulation outputs via Tucker decomposition, product-Grassmann interpolation, Procrustes core alignment, and a Core-of-Cores low-rank surrogate. Achieves $35.6 \times 564 \times$ complexity reduction over flattened baselines across four PDE benchmarks; the methodology extends naturally to biomedical image and sensor fields where spatial structure must be preserved. *Manuscript under review.*
- **Active-Subspace Gaussian-Process Emulation for Tensor-Valued Simulations.** Developing a surrogate framework for simulations with tensor-valued inputs and outputs, as a companion to the Tensor ROM work above. Reduces the input through supervised tensor active subspaces, represents the output via a higher-order decomposition, and places a decomposition-based Gaussian process on the reduced map to deliver field-level predictive uncertainty; an active-learning loop then selects new high-fidelity runs where predictive variance is largest. Establishes finite-sample subspace-estimation guarantees through matrix perturbation theory (Wedin / Davis–Kahan). *Manuscript targeting Technometrics.*

Amirkabir University of Technology

Research Assistant (Aerospace) — Prof. Mehdi Sabzeh Parvar

Tehran, Iran

Jan 2020 – Jan 2021

- Developed a 6-DOF flight dynamics simulator in MATLAB/Simulink and trained neural network models to estimate safe operating envelopes under hybrid control inputs; introduced machine learning-based safety analysis for nonlinear dynamical systems.

Leadership, Teaching, and Mentorship

Georgia Institute of Technology

Head Teaching Assistant — ISyE 6525: High-Dimensional Data Analytics (Prof. Paynabar) Jan 2024 – Present

Atlanta, GA

- Lead and coordinate a TA team for a 80+ student graduate course covering nonparametric methods, dimensionality reduction, sparse modeling, manifold learning, and high-dimensional regression. Serve as the primary point of contact for course operations, logistics, and student support.
- Design advanced problem sets and final projects, mentor student research projects on uncertainty quantification and interpretability, and conduct weekly office hours.
- Recognized as **Outstanding Online TA** (2025, Tech to Teaching Program, Georgia Institute of Technology).

Royan Institute

Course Instructor — Artificial Intelligence Laboratory

Tehran, Iran

Aug 2022 – Aug 2023

- Designed and delivered a project-based AI curriculum covering rule-based expert systems, hybrid architectures, and applied machine learning in clinical healthcare contexts. Supervised end-to-end student projects from problem formulation through evaluation.

Amirkabir University of Technology

Teaching Assistant — Principles of Simulation (Prof. Abbas Ahmadi)

Tehran, Iran

Sep 2022 – Aug 2023

- Provided instruction in Python, MATLAB, Simulink, Arena, and AnyLogic for stochastic simulation and computational modeling, including healthcare operations simulation. Graded assignments and examinations.

Amirkabir University of Technology

Teaching Assistant — Engineering Dynamics (Prof. Hamid Reza Ovesy)

Tehran, Iran

Feb 2020 – Aug 2023

- Delivered weekly recitations on analytical mechanics, Lagrangian dynamics, and stability analysis. Led office hours and graded assignments and examinations.
- **Ranked #1 Teaching Assistant in the department** by student evaluation for three consecutive years.

Selected Projects in Health Systems and Analytics

- **Nurse Scheduling with Quick-Response Methods:** Formulated and solved mixed-integer programming models for healthcare workforce scheduling under demand uncertainty. *GAMS* (2020–2021)
- **Breast Cancer Prediction Benchmark:** Evaluated and compared machine learning classifiers for disease prediction with emphasis on statistical validity, calibration, and generalization. *Python* (2021–2022)
- **Healthcare Operations Quality Improvement:** Applied Six Sigma methodology to identify and reduce service quality deficiencies in a healthcare organization; delivered managerial recommendations on process standardization. (2021–2022)
- **Food Recommender Expert System:** Designed a knowledge-based expert system tailored to individual user health characteristics and dietary needs. *Python* (2021–2022)
- **Anomaly Detection via Sliding Window Control Charts:** ARMA-based statistical monitoring with sliding window control charts for early anomaly detection in sensor streams. *Python* (2023)
- **Sensitivity-Guided Diffusion Models for Scientific Dynamical Systems:** Diffusion-based generative models constrained to physically meaningful latent subspaces via sensitivity-informed perturbation schedules; applicable to data-scarce scientific and biomedical settings. *Python, PyTorch* (2024–2025)

Honors and Awards

- **Outstanding Online TA Award**, ISyE 6525, Tech to Teaching Program, Georgia Institute of Technology 2025
- **Stewart Topper Fellowship**, Georgia Institute of Technology 2023
- **Mansour Doctoral Fellowship**, North Carolina State University — granted for exceptional academic promise in Industrial Engineering 2023
- Direct entry to M.S. in Industrial Engineering (Healthcare Systems) as an elite student, ranked **3rd of 120** applicants 2021
- Direct entry to M.S. in Aerospace Engineering as an elite student, ranked **2nd of 120** applicants 2021

- Selected as an elite student and approved to pursue a dual degree in Aerospace Engineering 2018
- Ranked in the **top 1%** among 150,000+ participants in the National University Entrance Exam (Physics & Mathematics) 2016

Technical Skills

- **Programming and computing:** Python, PyTorch, NumPy, SciPy, pandas, scikit-learn, R, MATLAB, SQL, Git, Linux, L^AT_EX; high-performance computing (SLURM, GPU clusters)
- **Reduced-order and surrogate modeling:** geometry-aware reduced-order modeling, tensor decompositions (Tucker, HOSVD), product-Grassmann interpolation, active subspaces, Gaussian-process emulation, design of experiments (fractional factorial, RSM)
- **Uncertainty quantification and trustworthy ML:** conformal prediction, selective prediction, calibration, sensitivity analysis, decision-making under missing modalities and distribution shift
- **Machine learning:** deep learning, gradient boosting (XGBoost), Gaussian processes, representation and multimodal learning, diffusion generative models
- **Simulation and operations research:** AnyLogic (agent-based + discrete-event), Arena, Simulink, GAMS; mixed-integer programming; healthcare operations simulation
- **Mathematics and theory:** high-dimensional statistical inference, optimization on Riemannian manifolds, matrix perturbation theory (Wedin / Davis–Kahan), tensor algebra, stochastic processes

Service and Outreach

- Core team member, CANSAT satellite design competition, Amirkabir University of Technology 2019
- Staff organizer, National AerUp Competition, Amirkabir University of Technology 2017, 2018
- Referee, National Aerospace Competitions, Aerospace Engineering Department, Amirkabir University of Technology 2017, 2018

Languages

- **Persian:** Native
- **English:** Excellent (TOEFL iBT 110 / 120: Reading 28, Listening 30, Speaking 25, Writing 27)
- **Korean:** Excellent spoken proficiency